MILITARY SPECIFICATION

RELAY, MINATURE 6 PDT

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Minature Relay which is used in the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification, Marking of U.S. Military Property.
MIL-STD-456	- Electronic Parts, Date and Source, Coding For.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

U.S. ARMY

9237024 - Relay Miniature (6 PDT.)

FSC 1190

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The relay shall comply with all requirements specified on the drawing (dwg)9237024, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Enclosure seal. The enclosure seal leakage rate of the relay shall not exceed 1 X 10^{-8} atm. c.c./sec (air equivalant) when tested as specified in 4.4.1.

3.7 Insulation resistance. The insulation resistance of each circuit when measured against all other circuits and case shall be 1000 megohms minimum, when 500 + 50 vdc has been applied with the current limited to 5 milliamperes and tested as specified in 4.4.2.

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From	To
1	2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 19, case
2	3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 19, case
3	4, 5, 6, 7, 8, 9, 10, 11, 12,19, case
4	5, 6, 7, 8, 9, 10, 11, 12, 19, case
5	6, 7, 8, 9, 10, 11, 12, 19, case
6	7, 8, 9, 10, 11, 12, 19, case
7	8, 9, 10, 11, 12, 19, case
8	9, 10, 11, 12, 19, case
9	10, 11, 12, 19, case
10	11, 12, 19, case
11	12, 19, case
12	19, case
19	case

3.8 Dielectric strength. The dielectric strength of the circuits listed in Table I shall withstand a potential of 1000 volts root mean square (vrms) 60 + 5 Hertz (Hz) and the circuits listed in Table II shall withstand 500 vrms, 60 + 5 Hz, when applied for 15 seconds minimum without evidence of insulation breakdown arcing or flashover when tested as specified in 4.4.3.

Table I

From	To
1	3, 5, 7, 9, 11, case
3	5, 7, 9, 11, case
5	7, 9, 11, case
7	9, 11, case
9	11, case
11	case

Table II

From	To
2	4, 6, 8, 10, 12, 20, case
4	6, 8, 10, 12, 20, case
6	8, 10, 12, 20, case
8	10, 12, 20, case
10	12, 20, case
12	20, case
20	case.

3.9 Contact circuit resistance. The contact resistance shall be 0.050 ohms, maximum when measured at the terminals when tested as specified in 4.4.4. Record actual contact resistance in accordance with 3.18 between terminals (1 and 2) and (3 and 4), code 070001; (5 and 6), code 070002; (7 and 8), code 070003; (9 and 10), code 070004; (11 and 12), code 070005.

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3.10 Pickup voltage. The relay pickup voltage shall be 18.0 vdc maximum when it is indicated that the normally closed contact is open and all normally open contacts are closed when tested as specified in 4.4.5. Record actual pickup voltage required in accordance with 3.18 code 080001.

3.11 Dropout voltage. The relay dropout voltage shall be greater than 1.4 vdc and less than 14.1 vdc when it is indicated that all closed, normally open, contacts have been opened and the opened, normally closed, contacts have closed when tested as specified in 4.4.5. Record actual dropout voltage in accordance with 3.18 code 090001.

3.12 Operate, release and bounce time. The relay shall operate in 0.010 second maximum, excluding bounce time, when it is indicated that all normally closed contacts have opened, and all normally open contacts have closed. The coil energize voltage shall be 26 ± 5 vdc, except when otherwise specified. The contact current through each contact shall be 0.1 ± 0.05 amperes when 27 vdc maximum is applied in series with a 50 ohm minimum resistor which will produce the current required. The bounce time of each contact pair shall not exceed 0.001 seconds under these conditions, when tested as specified in 4.4.6. The release time shall be 0.010 second maximum. Record in accordance with 3.18; actual pickup time code 100001, actual bounce time code 100002 and actual dropout time code 100003.

3.13 D.C. Coil resistance. The direct currect (dc) resistance of the relay coil shall be 200 ± 20 ohms at 25 ± 3°C when tested as specified in 4.4.7. Record coil resistance in accordance with 3.18 code 110001.

3.14 Contact chatter. Contact chatter shall not exceed 0.000010 seconds duration when the relay is in energized and de-energized condition using a contact load of 0.002 ± 0.005 amp when tested as specified in 4.4.8. Record actual values in accordance with 3.18.

3.15 Low level switching life. The relay shall be able to withstand 100,000 operations, have no phase-to-phase arcing, no failure to operate, and no failure to release. The opening or closing of all contacts shall be continuously monitored. The relay shall comply with the requirement of 3.7 through 3.13 after testing as specified in 4.4.9. Record actual values in accordance with 3.18.

Requirements	Test Code No.
Insulation resistance (3.7)	013001
Dielectric strength (3.8)	013002
Contact resistance (3.9)	013003 thru 013008
Pickup voltage (3.10)	013009
Dropout voltage (3.11)	013010
Release, operate, contact time (3.12)	013011 thru 013013
Coil resistance (3.13)	013014

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3.16 Environmental testing. The relay shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The relay shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.10. Tests shall be conducted in the order specified.

3.16.1 Temperature shock. The relay shall withstand temperature shocks between +160° ± 5°F to -65° ± 5°F for five (5) complete cycles, without evidence of external damage (see 6.5) and shall comply with the requirements of 3.7, 3.10 and 3.12 during temperature extremes and of 3.8 after testing as specified in 4.4.10.1.

Record actual values in accordance with 3.18.

Requirements	Test Code No (During)	
	Lo Temp	Hi Temp
Insulation resistance (3.7)	014001	114001
Pickup voltage (3.10)	014002	114002
Dropout voltage (3.11)	014003	114003
Release, operate, contact time (3.12)	014004,5,6	114004,5,6
	(After)	
Dielectric strength (3.8)	214001	

3.16.2 Transportation vibration. The relay shall withstand vibration tests along each of three (3) mutually perpendicular axes as designed in the table below without evidence of external damage (see 6.5) and shall be tested as specified in 4.4.10.2. Frequency shall be cycled at a logarithmic rate between 5-500-5 Hertz (Hz) in fifteen (15) minutes. At the completion of this test, the relay shall comply with 3.7, 3.9, 3.10 and 3.11.

Vibration Test Schedule

Frequency (Hz)	Amplitude	Sweep Cycles		
		Ambient	-60°F	+160°F
5-15	0.4" da	3	1	1
15-80	5 g	3	1	1
80-115	0.015" da	3	1	1
115-500	10 g	3	1	1

Record actual values in accordance to 3.18.

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Requirements	Test Code No. Amb	Lo Temp	Hi Temp
Insulation resistance (3.7)	015001	115001	215001
Combat resistance (3.9)	015002-8	115002-8	215002-8
Pickup voltage (3.10)	015008	115009	115009
Dropout voltage (3.11)	015010	115010	215010

3.16.3 Operational vibration. The relay shall be vibrated at room temperature -60° ± 5°F, and +160 ± 5°F in each of three mutually perpendicular axes as specified below without evidence of external damage (see 6.5) and shall be tested as specified in 4.4.10.3. Contact chatter (see 3.14) shall be monitored during operational vibration. After testing, the relay shall comply with 3.7, 3.9, 3.10 and 3.11.

(a) 0.06 g 2/Hertz (Hz) power spectral density 9.3 rms g's minimum at frequencies ranging from 100 to 1000 Hz, attenuated 12 decibels per octave so that the total frequency enveloped is 50 to 2000 Hz for 10 seconds. The tolerance level shall be ± 3 decibels (db) between 100 to 1000 Hz and ± 6 db between 1000 to 2000 Hz.

(b) 0.02 g 2/Hz power spectral density (5.3 rms g's minimum) at frequencies ranging from 100 to 1000 Hz, attenuated 12 db so that the frequency enveloped is 50 to 2000 Hz for one (1) minute. The tolerance level shall be ± 3 db between 100 to 2000 Hz.

Record actual values in accordance with 3.18.

Requirements	Test Code No.
Chatter (3.14)	016001
Insulation resistance (3.7)	116001
Contact resistance (3.9)	116002 thru 008
Pickup voltage (3.10)	116009
Dropout voltage (3.11)	116010

3.16.4 Mechanical shock. The relay shall withstand a force of 50 ± 5 gravity units (g's) peak amplitude, three (3) times in each direction along its three (3) mutually perpendicular axes (total 18 shocks), without evidence of external damage (see 6.5) when tested as specified in 4.4.10.4. The time duration of the half sine wave shock shall be 11 ± 1.0 milliseconds (ms). The relays shall comply with the requirements of 3.14 during testing and 3.7, 3.9, 3.10 and 3.11 after testing.

Record actual values in accordance with 3.18.

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Requirements	Test Code No.
Chatter (3.14)	017001
Insulation resistance (3.7)	117001
Contact resistance (3.9)	117002 thru 008
Pickup voltage (3.10)	117009
Dropout voltage (3.11)	117010

3.16.5 Acceleration and spin. The relay shall withstand the loading imposed by the following simultaneous levels of acceleration for a period of three minutes minimum in each direction along each of its three mutually perpendicular axes. In each direction, the coil shall be de-energized for three minutes, and rated coil voltage (or current) shall be momentarily applied; the voltage (or current) shall be reduced to one-half rated for an additional three minutes. Contact chatter (3.14) shall be monitored for proper position during the test as specified in 4.4.10.5.

a. Constant acceleration of 30 g's (maximum).
b. Centripetal acceleration resulting from a constant rotational speed of 300 rpm with the longitudinal axis of the relay located 8 inches from the center of rotation.

The relay shall comply with the requirements of 3.12 during testing and 3.7, 3.9, 3.10 and 3.11 after testing.

Record actual values in accordance with 3.19.

Requirements	Test Code No.
Chatter (3.14)	018001
Insulation resistance (3.7)	118001
Contact resistance (3.9)	118002
Pickup voltage (3.10)	118003
Dropout voltage (3.11)	118004

3.17 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460. In addition, prior to potting all soldered assemblies shall comply with the requirements of MIL-STD-1460.

3.18 Test codes (see 3.5). The contractor shall provide data feedback on punch cards in conformance with Qaulity Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.19 Marking. Relays shall be marked in accordance with MIL-STD-130 with the following information:

a. Manufacturers name or trademark.
b. Government dwg number, type designation,and serial number (see 3.5).

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c. Date code in accordance with MIL-STD-456.
d. Switch terminals shall be marked as specified in the applicable detail specification or dwg.

3.20 Workmanship.
3.20.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.
3.20.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.
3.20.3 Burrs. Parts shall be free from burrs.
3.20.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.
3.20.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.
4.2 First article inspection.
4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies (see dwg 9237024) for first article testing to the testing facility designated in the invitation for bids or request for proposal.
4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

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Table III
First article inspection and tests

Inspection or Test	Dest	Non Dest	Unit Identification 1	2	3	4	5	6	7	8
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X	X	X	X	X	X
2. Temperature shock (see 3.16.1)	X			X	X		X			X
3. Transportation vibration (see 3.16.2)	X			X		X		X		X
4. Mechanical shock (see 3.16.4)	X				X	X			X	X
5. Acceleration and spin (see 3.16.5)	X						X	X	X	X
6. Operational vibration and temperature (see 3.16.3)	X		X	X	X	X	X	X	X	X
7. Insulation resistance (see 3.7)		X	X	X	X	X	X	X	X	X
8. Dielectric strength (see 3.8)		X	X	X	X	X	X	X	X	X
9. Contact resistance (see 3.9)		X	X	X	X	X	X	X	X	X
10. Pickup voltage (see 3.10)		X	X	X	X	X	X	X	X	X
11. Drop out voltage (see 3.11)		X	X	X	X	X	X	X	X	X
12. Operate, release, bounce time (see 3.12)		X	X	X	X	X	X	X	X	X
13. DC coil resistance (see 3.13)		X	X	X	X	X	X	X	X	X
14. Seal test (see 3.6)		X	X	X	X	X	X	X	X	X
15. Life cycling (see 3.15)	X		X		X			X		X

Tests 7 through 14 need not be in the order specified.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer or raw material or components, one production method or manufacturing operation. The entire lot shall be submitted as a unit for inspection.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Relay, minature (6PDT) (see dwg 9237024).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Hole diameter, incorrect	Gage
	102. Hole centerline distance incorrect	Gage
	103. Flange to terminal end dimension, incorrect	Gage
	104. Terminal extension, long length, incorrect	Gage
	105. Terminal extension, short length, incorrect	Gage
	106. Blue bead missing	Visual
	107. Bent or damaged terminals	Visual
	108. Marking missing, incorrect, incomplete or illegible	Visual
	109. Workmanship (see 3.20)	Visual
	110. Objective evidence of inspection for solder (see 3.17)	Visual
	111. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Serialization improper or missing (see 3.4)	Visual

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4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.20)	Visual

4.3.3 Testing. All tests shall be performed. Tests need not be tested in the order specified below:

	Test	Test Classification
a.	Insulation resistance	Major
b.	Dielectric strength	Major
c.	Contact resistance	Major
d.	Pickup voltage	Major
e.	Drop out voltage	Major
f.	Operate, release,bounce time	Major
g.	Direct current coil resistance	Major
h.	Seal test	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Insulation resistance, dielectric strength, contact resistance, pickup voltage, drop out voltage, operate, release, bounce time, d.c. coil resistance and seal test. Each relay shall be tested in accordance with 4.4.1 through 4.4.7. If any item fails to comply with drawing requirements, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions designed or specified by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T as specified on DI-R-1714 (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. All tests are conducted at room ambient 80 ± 20°F unless specified otherwise.

4.4.1 Enclosure seal. The relay shall be placed in an approved fixture and utilizing a helium mass spectrometer leak detector measure the relay enclosure leak rate as specified in 3.6.

4.4.2 Insulation resistance. The insulation resistance shall be measured on approved equipment at a potential of 500 ± 50 volts direct current (vdc) and current shall not exceed 5 milliamperes (ma). The potential shall be applied for a minimum of 5 seconds to a maximum of 60 seconds while testing to determine compliance with 3.7.

4.4.3 Dielectric strength. The dielectric strength shall be measured with an approved tester while being tested to determine compliance with 3.8.

4.4.4 Contact circuit resistance. The relay shall be tested on an approved circuit resistance tester to determine compliance with 3.9. The magnitude of the current (dc) throught the contacts shall be 100 ± 10 milliamperes (ma) and the open circuit test voltage shall not exceed 32 vdc. Separate voltage and current hook-up wiring shall be used at each terminal in order not to include the effect of hook-up wiring on measurement values. Activation of relays for purpose of improving contact surfaces prior to actual measurement of resistance shall not be permitted.

4.4.5 Pickup voltage and dropout voltage. The pickup voltage and the dropout voltage shall be measured with approved equipment. The pickup voltage shall be increased in one volt steps. An oscillation effect produced at insufficient energize voltage causes a buzzing sound and is considered acceptable. Voltages shall comply with 3.10 and 3.11. Record values (see 3.10 and 3.11).

4.4.6 Operate, release and bounce times. Each relay shall be tested on approved equipment which will record pickup, dropout and bounce time with the specified voltages across coil and contact circuits. The relay shall comply with 3.12. Each normally open and normally closed contact of relay shall be tested. Results of all contact tests of a relay may be obtained simultaneously. Indicator used shall have a

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response time of 0.001 second maximum. Record actual values (see 3.12).

4.4.7 D.C. coil resistance. Each relay shall be tested on approved equipment which will measure the direct current dc resistance of the coil. Record actual value (see 3.13).

4.4.8 Contact chatter. Contact chatter shall be tested on approved equipment which will measure momentary contact open or closure (see 3.14) under environmental test conditions of operational vibration (3.16.3), mechanical shock (3.16.4), and acceleration spin (3.16.5).

4.4.9 Low level switching life. Relays shall be tested in accordance with Method 311 of MIL-STD-202. The following details shall apply:

(a) Temperature environment: -65°C to 125°C.
(b) No "misses" of relay contacts are allowed.
(c) Maximum contact resistance allowed is 0.100 ohms.
(d) Monitoring of contacts for "sticking" is applicable.
(e) Number of operation cycles is 100,000 and the cycling rate is two (2) operations per minute.

4.4.10 Environmental testing (see 3.16).

4.4.10.1 Temperature shock. The relay shall be placed in an approved temperature chamber stabilized at +160° ± 5°F (see 6.4) and the relay shall remain at this temperature for a minimum period of four hours. At the conclusion of this period, and within five minutes, the relay shall be transferred to another approved chamber stabilized at -60° ± 5°F (see 6.4). The relay shall remain at this temperature for a minimum period of four hours. This constitutes on cycle. The number of complete cycles shall be five. At the conclusion of the fifth cycle, the relay shall be removed from the test chamber, returned to room temperature and within a period of one hour test for compliance with 3.16.1.

4.4.10.2 Transportation vibration. The relay shall be mounted securely on a fixture capable of being positioned along each direction along its mutually perpendicular axes. The mounted relay shall then be installed in a suitable temperature vibration chamber set at -65° ± 5°F for 4 hours minimum. The relay shall then be vibrated as specified in 3.16.2. The relay shall then be prepared and tested above at a temperature of +160° ± 5°F. Remove the relay from the chamber, allow to return to room temperature, visually examine for external damage and test for compliance with 3.7, 3.9, 3.10 and 3.11.

4.4.10.3 Operational vibration. The relay shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted relay shall then be installed on a vibrator and vibrated to the levels and tests as specified in 3.16.3.

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4.4.10.4 Mechanical shock. The relay shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted relay shall then be installed on a shock producing device and tested to the levels and tests as specified in 3.16.4.

4.4.10.5 Acceleration and spin. The relay shall be mounted securely on a fixture capable of being readily positioned along each direction along its mutually perpendicular axes. The mounted relay shall then be installed on a centrifuge and accelerated to the levels and tests as specified in 3.16.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The relay shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The relay shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The relay shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.9 through 3.15, 3.16 through 3.16.5 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

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6.5 External damage. Characteristics such as obliterated markings, damaged finish exposing base metal, rust and cracks that are detectable by visual examination, shall be considered cause for rejection.

6.6 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-1196 dated 29 June 1971.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A098

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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